1. Transport in Animals

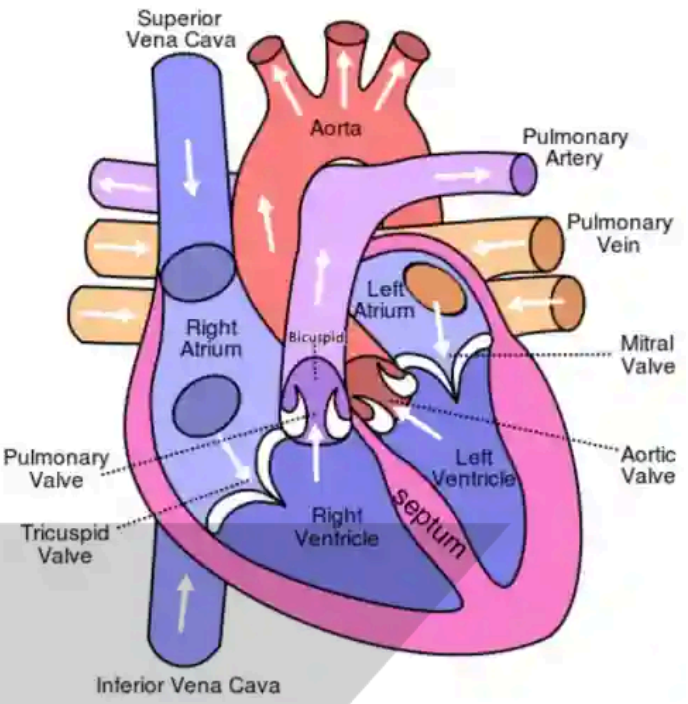
1.1. Circulatory Systems

Circulatory System: a system of tubes (veins, capillaries, arteries) with a pump (heart) and valves (in heart and veins) to ensure a one-way flow of blood.

- Single Circulation System (fish):
  - Blood flows through the heart once every complete circuit (No Septum)
  - Two heart chambers (Atrium and Ventricle)
  - Blood absorbs oxygen in the gills in lower-pressure
  - Released in body cells, then back to the heart
- Double Circulation System:
  - Four heart chambers
  - Blood passes through the heart twice every complete circuit
  - Oxygenated in the lungs, to the heart, to the body, and back to the heart
- Advantages:
  - Delivers a greater blood flow rate to tissues around the body as the heart pumps the rich oxygenated blood to it from the lungs,
  - Prevents the mixing of oxygenated and deoxygenated blood.
  - Lowers pressure to lungs - protects lung tissues from damage
  - More efficient - supports high metablic rates in warm-blooded animals like mammals and birds.

1.2. Heart

The mammalian heart contains a **systemic and pulmonary circuit**.



- Right Atrium: collect deoxygenated blood & pump it to the right ventricle
- Right Ventricle: pumps deoxygenated blood to lungs
- Pulmonary Artery: carries deoxygenated blood from the right ventricle to the lungs
- Septum: separates the left and right sides of the heart and keeps deoxygenated and oxygenated blood separate.
- Pulmonary Vein: carries oxygenated blood from the lungs to the left atrium
- Left Atrium: collect oxygenated blood and pump it to the left ventricle
- Left Ventricle: pumps oxygenated blood to the body via the aorta
- Aorta: carries oxygenated blood from the left ventricle to the rest of the body
- Atrioventricular and Semi-lunar Valves: prevent backflow of blood

IMPORTANT! Relative Muscle Wall Thickness: Atria < Right Ventricle < Left Ventricle

1.4. Functioning of the Heart

- The deoxygenated blood incoming from the body flows into the **right atrium** via the **vena cava.**
- Once the right atrium has filled with blood, the blood is pushed through the **atrioventricular valve** into the **right ventricle.**
- The ventricle contracts and the blood is pushed into the **pulmonary artery** through the **semilunar valve,** which prevents blood from flowing back into the heart.
- The blood travels to the lungs and moves through the capillaries, passing the alveoli.
- **Oxygenated blood** returns to the **left atrium** via the **pulmonary vein**

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- It passes through the **atrioventricular valve** into the **left ventricle**
- The thicker muscle walls of the ventricle contract to push the blood forcefully into the **aorta** and around the body
- The **semilunar valve** in the aorta prevents the blood from flowing back down into the heart

Example Past Year Question

Explain the reasons for changes in pressure seen in arteries (0610/42/F/M/23)

- caused by contraction of muscles (of the heart/ventricle)
- pressure increases when the heart / ventricles contract/pump
- pressure decreases when the heart/ventricles relax

1.5. Exercise on Heart Rate

- The heart's electrical activity can be monitored by the electrocardiogram (ECG), pulse rate, stethoscope and listening to the sounds of the valves closing.
- Physical activity makes the heart beat more quickly and deeply for increased blood circulation so that more oxygen and glucose can get to the muscle.

1.6. Coronary Heart Disease

- The coronary arteries are the **heart's blood supply**.
- They supply oxygenated blood and nutrients, allowing it to continue contracting and pumping blood.
- The coronary artery becomes **blocked**, interrupting blood supply to the heart muscle.
- Part of the heart muscle stops contracting, causing a heart attack
- Risk factors are diet, lack of exercise, stress, smoking, genetic predisposition, age and sex
- This can be prevented by not smoking, avoiding fatty food (a good diet) and exercising regularly

1.7. Blood Vessels

Vessel	Function	Structure
Arteries	Transport high-pressure blood away from heart	Elastic tissue walls stretch and relax as blood is forced out; causes pulse
		Thick walls to withstand high pressure
		Small lumen maintains (high) blood pressure.
Veins	Transport low pressure blood to the heart	Valves prevent backflow of blood.
		Blood is at low pressure, but nearby muscles squeeze veins and help push blood to the heart
		Large and wide lumen to reduce resistance to the flow of blood
Capillaries	Allow substances to diffuse into cells	One cell thick wall for easy diffusion
		Highly branched; large surface area
		Capillary beds constantly supplied with fresh blood, so diffusion occurs

Major Blood Vessels

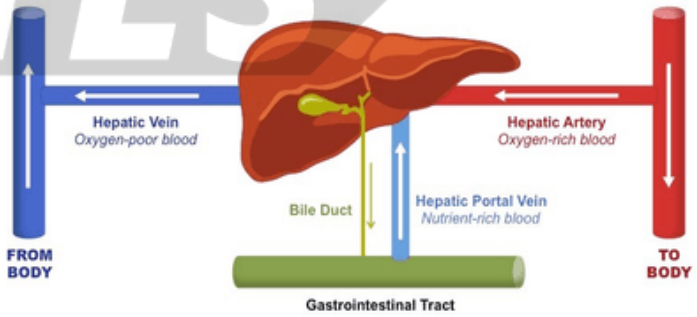
Heart: Vena Cava, Aorta, Pulmonary Arteries & Vein

Lungs: Pulmonary Arteries and veins

Kidney: Renal Arteries and veins

Liver: Hepatic Artery, Hepatic Veins and Hepatic Portal Vein

- The **hepatic artery** brings oxygenated blood from the heart to the liver
- The **hepatic vein** brings deoxygenated blood from the liver back to the heart
- The **hepatic portal vein** carries blood from the gastrointestinal tract, gallbladder, pancreas and spleen to the liver. This blood contains nutrients and toxins extracted from digested contents.



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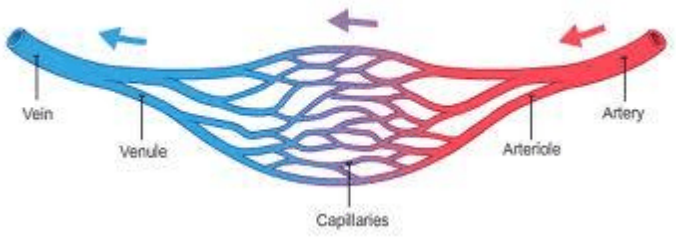


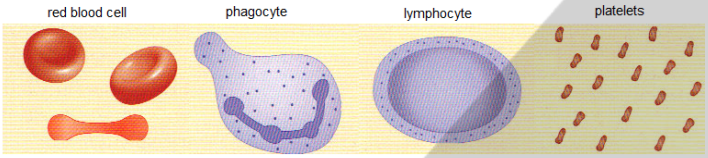
Arterioles and Venules

- The vessels that connect arteries to capillaries are called **arterioles**
- The vessels that connect capillaries to veins are called **venules**

1.8. Blood

- **Red blood cells**: biconcave shape, haemoglobin and oxygen transport (oxy-haemoglobin)
- **White blood cells**: phagocytosis and antibody production
- **Platelets**: allows/promotes blood clotting
- **Plasma**: transport of blood cells, ions, nutrients, urea, hormones and carbon dioxide (mostly water and dissolved substances)

> **Syllabus 9.4.2**: You must be able to identify red and white blood cells in photomicrographs and diagrams



White Blood Cells

Phagocyte	Lymphocyte
Phagocytes have lobed/irregular C-shaped nuclei and vesicles containing digestive enzymes.	Lymphocytes have a circular nucleus and are found in blood
Phagocytosis: engulfs pathogen, vesicles fuse with the vacuole, enzymes digest bacteria.	Large nucleus/small cytoplasm, and they produce antibodies,
Antigens: protein/carbohydrate on the surface of the pathogen which provokes the immune system	**Antibodies**: Y-shaped proteins bind to label pathogens, marking them for destruction by the phagocytes.

Blood Clotting

- Reduces blood loss and keeps pathogens out
- Fibrinogen (inactive) turns to fibrin (activated), forms a mesh to trap red blood cells, and eventually dries to form a scab.

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